

Gathering requirements

Brainstorming ⚡⚡⚡



Brainstorming: A group activity designed to *generate a large number of diverse and creative ideas* for the *solution* of a problem.

In a brainstorming session there are no "bad" ideas. The aim is to *generate ideas; not to validate them*. Brainstorming encourages you to "think outside the box" and put "crazy" ideas on the table without fear of rejection.

User Surveys ⚡⚡⚡

Surveys can be used to *solicit responses and opinions* from a large number of stakeholders regarding a current product or a new product.

Observation ⚡⚡⚡

Observing users in their natural work environment can uncover product requirements. Usage data of an existing system can also be used to gather information about how an existing system is being used, which can help in building a better replacement e.g. to find the situations where the user makes mistakes when using the current system.

Interviews ⚡⚡⚡

Interviewing stakeholders and domain experts can produce useful information about project requirements.

Focus Groups ⚡⚡⚡

Focus groups are a kind of informal interview within an interactive group setting. A group of people (e.g. potential users, beta testers) are asked about their understanding of a specific issue, process, product, advertisement, etc.

Prototyping ⚡⚡⚡



Prototype: A prototype is a *mock up, a scaled down version, or a partial system constructed*

- to get users' feedback.
- to validate a technical concept (a "proof-of-concept" prototype).
- to give a preview of what is to come, or to compare multiple alternatives on a small scale before committing fully to one alternative.
- for early field-testing under controlled conditions.

Prototyping can uncover requirements, in particular, those related to *how users interact with the system*. UI prototypes or *mock ups* are often used in brainstorming sessions, or in meetings with the users to get quick feedback from them.

 A mock up (also called a *wireframe* diagram) of a dialog box:

Name: _____

Modifiers: ☒ public ☐ default ☐ private ☐ protected
☐ abstract ☐ final ☐ static

Superclass:

[source: plantuml.com]

💡 Prototyping can be used for *discovering* as well as *specifying* requirements e.g. a UI prototype can serve as a specification of what to build.

Product Surveys ⚡⚡⚡

Studying existing products can unearth shortcomings of existing solutions that can be addressed by a new product. Product manuals and other forms of documentation of an existing system can tell us how the existing solutions work.

 When developing a game for a mobile device, a look at a similar PC game can give insight into the kind of features and interactions the mobile game can offer.